Cover Letter — Information Sciences

14 July 2026

Dear Editors of *Information Sciences*,

On behalf of my co-authors, I am pleased to submit our original research article, **“Beyond IID Data: A Proxy-Based Computational Study of Quantum Oracle Sketching Robustness under Structured Non-IID Streaming”**, for consideration in *Information Sciences*.

What the paper does. Quantum oracle sketching has recently been shown to deliver exponential memory advantages for streaming classification, with a theoretical extension to correlated data. Our paper is the first systematic computational study of how that robustness behaves under the structured correlation families that real streams actually exhibit. We introduce operational correlation proxies (refreshing time τ and repetition number r), evaluate five synthetic non-IID regimes against three classical memory-bounded baselines (10 seeds, 95% CIs, paired Wilcoxon tests), and distil the results into a (τ, r) sensitivity landscape. Key findings include analytic proxies overestimating empirical refreshing time by up to 125× for long-memory data, and a proxy-model advantage crossover at Markov correlation $\rho \approx 0.88$ ($p = 0.037$). A two-dataset real-stream check (NYC Taxi and Machine Temperature, Numenta Anomaly Benchmark) places real streams on opposite sides of the favourable region using the same estimators.

Fit to Information Sciences. The paper sits squarely in the journal’s applied-informatics scope: memory-bounded streaming learning under non-stationarity. It directly engages recent *Information Sciences* work on non-stationary streaming — elastic online deep learning (Su et al., 2024), bi-dynamic-distribution online learning (Yan et al., 2024), and evolving-prototype stream classifiers (Wu et al., 2024) — and complements that line with a forward-looking quantum-streaming perspective grounded in classical baselines.

Declarations. This manuscript is original, has not been published, and is not under consideration elsewhere. All authors have approved the submission. We declare no competing interests and no external funding. A generative-AI-assistance declaration is included in the manuscript per Elsevier policy. All code and data are openly available (Zenodo DOI: 10.5281/zenodo.19831893; MIT licence), and every figure and table regenerates from seeded scripts.

Suggested reviewers: [to be entered by the authors in Editorial Manager].

Thank you for considering our work.

Sincerely, on behalf of all authors,

AbdelMoniem Helmy (corresponding author) Department of Information Systems and Technology, Faculty of Graduate Studies for Statistical Research, Cairo University, Cairo, Egypt abdelmoniem.hafez@cu.edu.eg · ORCID 0000-0001-5996-6019

Co-authors: Mohammed Farsi (Taibah University, Yanbu, Saudi Arabia); Muhammad Mahmoud (Matrouh University, Egypt)